Problem set 5
Due Thursday February 26 at 11:59pm

Exercise 5.1 (8 points). Use the definition of convergence to show that $\lim_{n\to\infty} \frac{2n+1}{3n-1} = \frac{2}{3}$.

Exercise 5.2 (8 points). Suppose (p_n) is a sequence in a metric space X that converges to some $q \in X$. Show that the set $\{p_n \mid n \in \mathbb{N}\} \cup \{q\}$ is compact.

Exercise 5.3 (10 points; Squeeze Theorem). Let (a_n), (s_n) and (b_n) be sequences of real numbers such that $a_n \leq s_n \leq b_n$ for all $n \in \mathbb{N}$. Prove that if $x_n \to t$ and $y_n \to t$, then $s_n \to t$.

Exercise 5.4 (10 points). Consider two possible norms on $\mathbb{R}^2$: for a vector $v = (v_1, v_2)$,

$$\|v\|_2 = \sqrt{v_1^2 + v_2^2} \quad , \qquad \|v\|_1 = |v_1| + |v_2| \, .$$

Then let

$$d_2(x,y) = \|x-y\|_2 \quad , \qquad d_1(x,y) = \|x-y\|_1 \, .$$

Either d_1 or d_2 makes $\mathbb{R}^2$ into a metric space (you need not prove this).

Now, suppose (x_n) is a sequence in $\mathbb{R}^2$. Prove that $x_n \to x$ in the metric d_2 if and only if $x_n \to x$ in the metric d_1. (Hint: first prove some inequalities relating d_1 to d_2.)

Exercise 5.5 (not for credit; extra practice). Suppose $E \subset X$. We say E is *sequentially closed* if every sequence (p_n) in E which is convergent in X is actually convergent in E. Prove that E is closed if and only if it is sequentially closed.

Exercise 5.6 (not for credit; extra practice). (This is a modification of Rudin's exercise 2.26) We say a metric space X is *sequentially compact* if every sequence in X has a convergent subsequence. In class we will show that compactness implies sequential compactness. Here we show the converse.

(1) A metric space is *separable* if it admits a countable dense subset. Show that if X is sequentially compact, it is separable.

(2) A *base* for a metric space X is a family of open sets $\mathcal{U} = \{U_\alpha\}$ such that for every open set V, and every point $x \in V$, there is an element of the family U_α such that $x \in U_\alpha \subset V$. Equivalently, every open set of X can be expressed as a union of a subfamily of $\mathcal{U}$. Show that if X is separable, it admits a countable base.

(3) Show that if X admits a countable base, than every open cover of X has a countable subcover.

(4) Show that if $F_1 \supset F_2 \supset \cdots$ is a nested sequence of nonempty closed subsets of a sequentially compact metric space X, then $\bigcap_{k\in\mathbb{N}} F_k$ is nonempty.

(5) Show that every countable cover of a sequentially compact metric space has a finite subcover. Hint: Suppose $\{G_i\}_{i\in\mathbb{N}}$ is a countable cover with no finite subcover, and consider the subsets $(G_1 \cup \cdots \cup G_k)^c$.